

A Falsification-Based Framework for Validating Structure in Time-Series Data

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Abstract

Detecting meaningful structure in time-series data remains challenging due to the prevalence of spurious correlations, non-robust patterns, and apparent dependencies that persist under generic similarity or statistical coincidence. We present a **falsification-first framework** for evaluating structured dependencies in time-series signals through a hierarchical system of three detectors targeting **time-order (n_1)**, **interaction (n_2)**, and **closure (n_4)** structure. Each detector operates under a fixed contract consisting of predefined observables, mandatory falsification tests, and strict decision criteria requiring systematic degradation under controlled structural disruption.

The framework is evaluated using instrument, falsification, robustness, reproducibility, and audit controls across multiple domains. Controlled validation confirms **fail-closed detector behavior**, preventing unsupported structural claims while preserving negative outcomes when falsification criteria are not satisfied.

Application to real datasets demonstrates distinct behavior across detector levels. The **n_1 detector** identifies directional asymmetry in gamma-ray burst data but rejects irreversibility when rarity under surrogate nulls is insufficient. The **n_2 detector** identifies alignment-dependent coordination through collapse under mismatch pairing, distinguishing interaction from generic similarity [7,8]. The **n_4 detector** evaluates pair-dependent structural validity, accepting closure only when internally consistent pairing degrades under mandatory closure-breaking controls.

Across industrial, environmental, grid-scale, and experimental datasets, both **PASS and FAIL outcomes** are explicitly retained under frozen detector contracts. Conventional measures such as correlation and coherence may remain elevated under mismatch-controlled conditions, whereas the proposed observable collapses, demonstrating its role as a complementary falsification-based validation layer rather than a replacement for existing methods.

Across all evaluations, structure is accepted only when it degrades correctly under its defining falsifier. The framework therefore provides a **reproducible and conservative validation system** for distinguishing constraint-dependent structure from patterns that persist under generic similarity, shared statistics, or global synchrony.

Keywords: time-series validation; falsification framework; interaction detection; closure structure; surrogate testing; reproducibility

1. Introduction

Identifying meaningful structure in time-series data remains a fundamental challenge across scientific domains. Signals arising in astrophysics, geophysics, engineering systems, and controlled laboratory experiments often exhibit complex patterns suggestive of underlying organization. However, distinguishing genuine structure from apparent structure produced by noise, smooth trends, preprocessing artifacts, or superficial similarity remains difficult. Many commonly used approaches rely on descriptive statistics, correlation-based measures, synchronization metrics, or model-dependent fitting procedures [2,6–9] that can identify patterns but do not determine whether those patterns are structurally meaningful under controlled disruption.

A central limitation of such approaches is the absence of explicit falsification. Signals may exhibit elevated correlation, coherence, synchrony, or other forms of regularity even when the underlying structure is non-robust or incidental. In particular, apparent structure may persist under transformations that should disrupt the defining relationship, including time reversal, surrogate randomization, perturbation, or misalignment [10]. Without systematic evaluation under such destructive conditions, it remains difficult to distinguish constraint-dependent organization from generic statistical persistence. As a result, conventional approaches may remain permissive, allowing false-positive structural interpretations to persist.

The approach proposed in this work addresses this limitation through a principle of **strict selectivity**. The framework is designed to reject weak, non-specific, or non-robust patterns, and therefore does not guarantee positive detection on arbitrary signals. Instead, negative outcomes are treated as meaningful results, reflecting the absence of sufficiently constrained structure under the defined falsification rules. This emphasis on controlled rejection ensures that accepted structure corresponds to reproducible and constraint-dependent behavior rather than generic similarity or shared statistical properties.

This work introduces a **falsification-based validation framework** for assessing structure in time-series data, in which structural interpretation is defined operationally through response to controlled disruption. Rather than relying on observable magnitude or persistence alone, the framework requires that detected behavior exhibit systematic degradation under transformations that violate its defining constraint. The detectors n_1 (time-order), n_2 (interaction), and n_4 (closure) implement this principle at progressively stricter levels, each enforcing rejection of structure that persists under its corresponding falsifier. The primary contribution is therefore not increased detection sensitivity, but a reproducible mechanism for rejecting unsupported structure and reducing false-positive structural interpretation.

The framework consists of three detectors, each targeting a distinct form of structure:

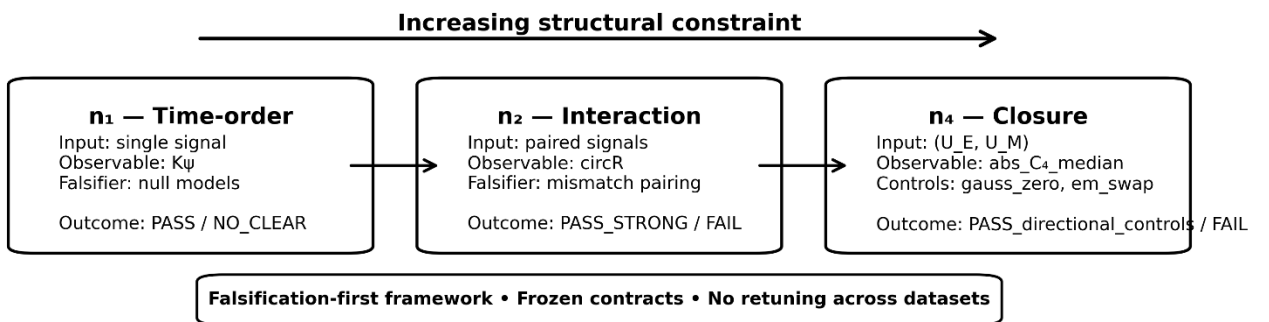
- **n_1 (time-order)**: evaluates directional time-order asymmetry under surrogate-based falsification without asserting full irreversibility [4,10];
- **n_2 (interaction)**: evaluates alignment-dependent coordination that collapses under mismatch pairing, distinguishing interaction from generic similarity [7,8];
- **n_4 (closure)**: evaluates pair-dependent structural validity by testing whether observed structure depends on internally consistent pairing and degrades under mandatory closure-breaking perturbations.

These detectors form a hierarchical ladder in which each level imposes progressively stricter structural constraints. The progression from **time-order** → **interaction** → **closure** reflects increasing specificity in the operational definition of structure and increasing reliance on falsification-based validation. Within this hierarchy, n_4 represents the strictest level, rejecting not only weak or non-robust patterns but also generic synchrony or coordination that does not satisfy internal structural consistency.

The framework is not intended to replace conventional statistical or signal-processing methods, but to complement them by evaluating whether detected structure remains valid under controlled falsification. Conventional similarity measures may remain elevated under mismatch or perturbation, whereas the proposed framework requires observable degradation under constraint violation. In this way, the framework provides a conservative and reproducible mechanism for reducing false-positive structural claims while preserving both positive and negative outcomes under fixed evaluation contracts.

Figure 1. Multi-level falsification framework for structure validation in time-series data.

The framework consists of three progressively stricter detector levels. The n_1 detector evaluates directional time-order asymmetry under surrogate null evaluation. The n_2 detector evaluates alignment-dependent coordination and requires collapse under mismatch pairing. The n_4 detector evaluates pair-dependent closure consistency and requires degradation under mandatory perturbation controls (gauss_zero, em_swap). Each detector operates under frozen observables, predefined falsifiers, and deterministic decision logic without retuning across datasets.



2. Results

2.1 Controlled Validation (Rössler)

To evaluate detector behavior under controlled and reproducible conditions, the framework was applied to the Rössler nonlinear dynamical system. The purpose of this analysis is not to establish new physical results, but to verify that the detectors behave consistently with their predefined structural contracts under a system with known and controllable dynamics. The Rössler system provides a stable validation environment in which structural regimes can be systematically varied through coupling while preserving reproducibility across runs.

The controlled evaluation focuses on three questions:

- whether detector responses vary systematically across dynamical regimes,
- whether mandatory falsifiers produce the expected collapse behavior,
- and whether outcomes remain consistent with the frozen detector contracts.

Across all evaluations, interpretation is determined not by observable magnitude alone, but by response to constraint-breaking transformations. Structure is therefore considered meaningful only when behavior observed under real conditions systematically degrades under the corresponding falsifiers.

n_1 — Time-Order Behavior

For single-channel trajectories, the n_1 detector produces measurable directional asymmetry under real conditions. Time reversal and surrogate constructions modify the directional statistic, confirming sensitivity to temporal ordering. However, separation from surrogate distributions does not consistently satisfy the frozen rarity criteria required for a positive classification. As a result, outcomes are predominantly **NO_CLEAR** under the strict n_1 contract. This behavior is consistent with the detector definition: the detector remains sensitive to directional structure while conservatively rejecting unsupported claims of irreversibility.

n_2 — Interaction Behavior

For paired signals derived from Rössler components, the n_2 detector evaluates alignment-dependent coordination across the coupling sweep. Under real pairing, measurable interaction behavior is observed throughout the sweep and strengthens systematically with increasing coupling. The strongest interaction behavior occurs within the synchronized regime, indicating that the observable tracks increasing coordination while remaining dependent on the mismatch-based falsification criterion defined in the detector contract.

Importantly, the detector does not operate as a generic synchrony measure. Interaction is interpreted as structurally meaningful only when alignment-dependent behavior collapses under mismatch pairing, ensuring that elevated coordination alone is insufficient for acceptance.

n_4 — Closure Behavior

For constructed paired representations, the n_4 detector evaluates pair-dependent structural validity through comparison between real pairing and mandatory perturbation controls. Under real conditions, closure deviation is measurable and systematically distinguishable from the perturbation controls (gauss_zero, em_swap), which produce consistent degradation.

Across the coupling sweep, closure-consistent behavior is concentrated within the weak-to-transition regime and progressively disappears as the system approaches global synchrony. This behavior indicates that closure is not a generic consequence of coordination, but depends on internally consistent pairing that is disrupted as the system collapses into a single coherent mode.

Accordingly, closure is interpreted as a selective structural property rather than a universal feature of synchronized systems.

Cross-Level Interpretation

A clear progression of structure is observed across coupling regimes. As coupling increases, baseline synchrony rises monotonically, reflecting transition from weak coordination to strong global synchrony. Within this progression:

- n_2 captures alignment-dependent coordination that strengthens with increasing coupling,
- whereas n_4 identifies selective structural organization concentrated prior to full synchronization and absent under global coherent collapse.

This distinction is shown in **Figure 2**, where the n_2 pass rate remains substantial across the sweep while n_4 structural pass behavior decreases and disappears as synchrony dominates. The results therefore demonstrate that closure is not a byproduct of synchronization, but a stricter structural condition requiring internally consistent pairing.

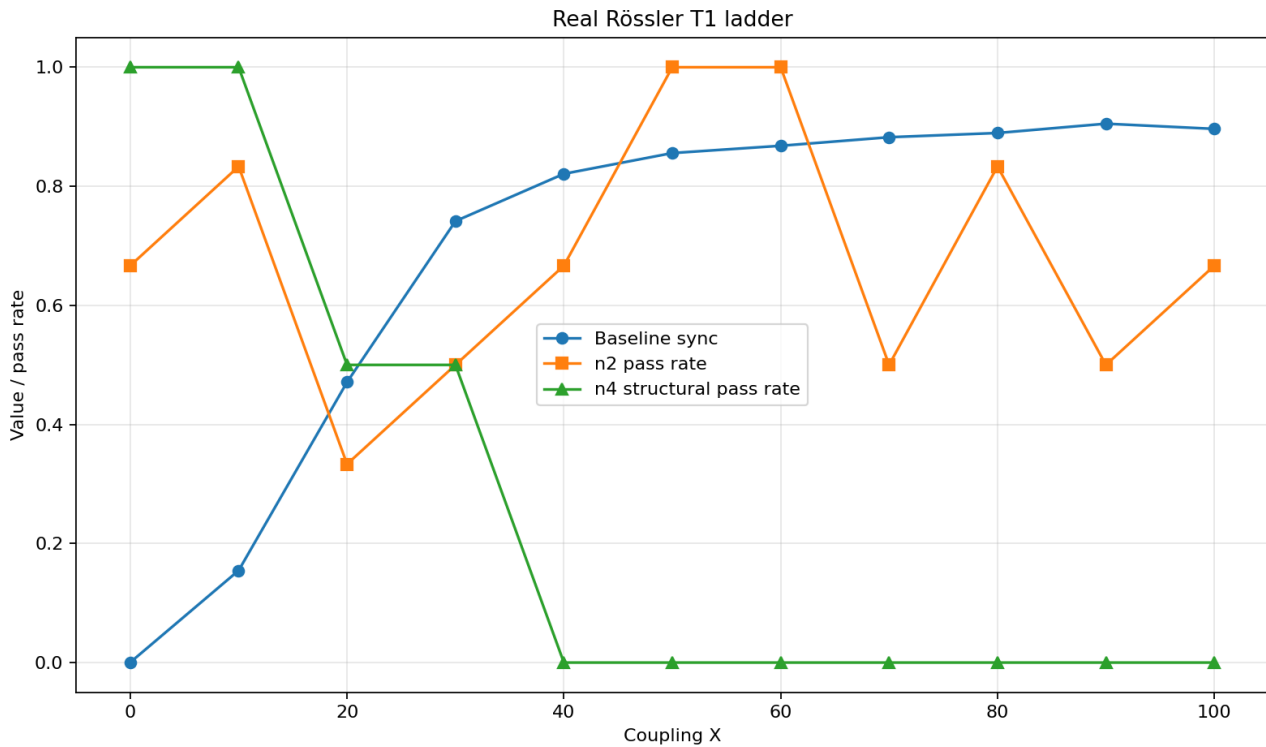
Across all detector levels:

- observables remain well-defined and measurable,
- falsifiers produce systematic and interpretable degradation,
- and outcomes remain consistent with the frozen detector contracts.

Importantly, negative outcomes are explicitly preserved, and no detector produces unsupported positive classifications under the controlled evaluation conditions.

Figure 2. Controlled validation of detector behavior across coupling in the Rössler system.

Baseline synchrony increases monotonically with coupling, reflecting transition from weak coordination to strong global synchrony. The n_2 pass rate remains substantial across the coupling sweep and is strongest in the synchronized regime, indicating alignment-dependent coordination. In contrast, n_4 structural pass behavior is concentrated in the weak-to-transition regime and disappears as synchrony dominates. This demonstrates that n_4 is not a generic synchrony detector, but identifies pair-dependent structural organization that exists prior to global coherent collapse.



2.2 n_1 — Time-Order Structure (GRB)

The first real-world evaluation applies the n_1 detector to gamma-ray burst (GRB) time-series data in order to assess whether directional temporal structure satisfies the strict falsification criteria defined by the framework. The analysis is performed under the frozen n_1 contract, which enforces fixed surrogate null families, mandatory time-reversal testing, deterministic execution conditions, and strict PASS / NO_CLEAR decision logic. Under this contract, structure is defined not by observable magnitude alone, but by statistical rarity under all required null families.

For representative GRB events, the detector produces a measurable directional time-order statistic under real conditions. Strong separation is observed relative to phase-randomized surrogates, indicating sensitivity to temporal organization. However, this separation does not remain statistically rare under block-shuffle nulls, preventing satisfaction of the full falsification criterion required for a positive classification. As a result, the final outcome remains **NO_CLEAR** under the strict n_1 contract.

Time-reversal evaluation further confirms directional sensitivity. Reversal systematically alters the directional statistic and produces consistent sign inversion, demonstrating that the observable responds to temporal direction. However, directional sensitivity alone is insufficient for structural validation. Under the framework definition, classification depends on rarity under all required falsifiers rather than reversal behavior alone.

This distinction is central to the interpretation of the result. The GRB signals exhibit measurable directional asymmetry, but that asymmetry is not uniquely time-order-dependent under the frozen falsification criteria. Similar behavior persists under ordering-disruptive null constructions, indicating that the detector correctly refuses to promote non-specific directional behavior to a claim of irreversibility.

Across repeated evaluations and multiple GRB events, the detector behavior remains stable and conservative:

- directional asymmetry is consistently observed,
- phase-surrogate separation persists,
- block-shuffle separation remains absent,
- and no artificial PASS outcomes are introduced.

The resulting **NO_CLEAR** classification therefore represents a meaningful validation outcome rather than a failure of the detector. The analysis demonstrates that the framework preserves negative outcomes when falsification support is incomplete, preventing false-positive claims of time-order structure.

2.3 n_2 — Interaction Structure (GOES)

The second-level evaluation applies the n_2 detector to paired time-series signals in order to test for **alignment-dependent coordination** under a strictly falsifiable protocol. The primary case uses GOES solar observation channels, treated as a paired system without imposing a physical coupling model. All evaluations are performed under the frozen n_2 contract, which enforces a fixed observable (circR), mandatory falsifiers (phase scramble, block shuffle, and mismatch pairing), deterministic execution conditions, and strict PASS_STRONG / FAIL classification logic.

Under the operational definition used in this framework, interaction is accepted only when the observed structure depends on correct temporal alignment and systematically collapses under

mismatch pairing. This distinction separates observable magnitude from falsification behavior and prevents interaction from being inferred solely from elevated similarity or coordination.

A controlled synthetic reference case first confirms detector capability under ideal alignment conditions. Under correct pairing, the detector produces elevated circR values together with systematic degradation under mismatch and randomized nulls, resulting in a PASS_STRONG outcome. This establishes that the observable responds correctly to alignment-dependent coordination under controlled conditions.

The detector was then applied to real GOES datasets. Under aligned conditions, measurable circR values are observed consistently across evaluated channels:

- GOES16: circR_real ≈ 0.154
- GOES15: circR_real ≈ 0.172

Under mismatch pairing, the observable collapses systematically:

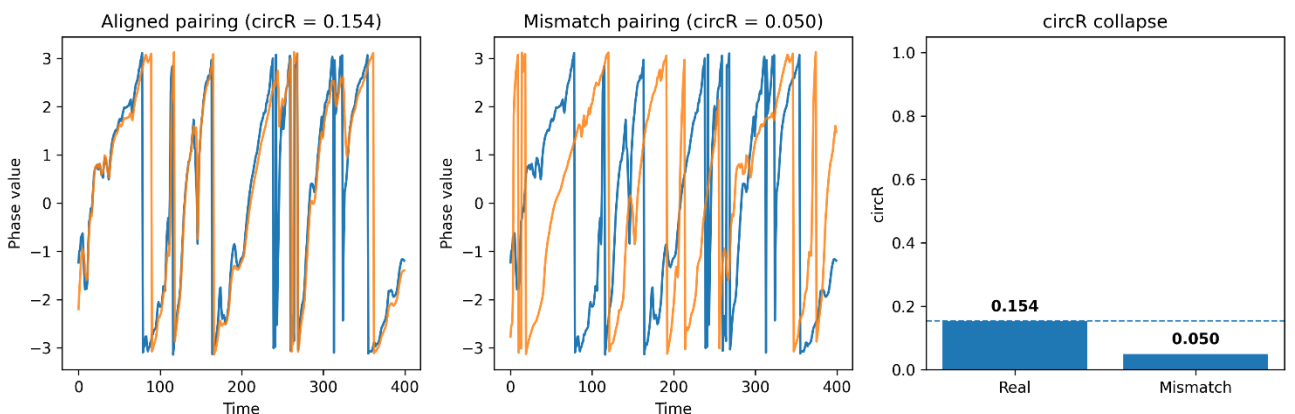
- GOES16 mismatch_mean ≈ 0.050
- GOES15 mismatch_mean ≈ 0.060

resulting in consistent **PASS_STRONG** classifications under the frozen detector contract.

This behavior is illustrated in **Figure 3**, where elevated coordination observed under correct alignment is reduced under mismatch pairing. The detector therefore accepts interaction only when the observable depends on preserved alignment and rejects coordination that persists independently of temporal correspondence.

Figure 3. Alignment-dependent coordination under correct and mismatch pairing.

The n_2 detector evaluates alignment-dependent coordination using the circR observable. Under correct temporal alignment, paired signals exhibit elevated circR values (≈ 0.154), whereas mismatch pairing disrupts temporal correspondence and reduces circR (≈ 0.050). Structure is accepted only when this collapse is observed under mismatch, demonstrating that the detected coordination depends on correct temporal alignment rather than intrinsic signal similarity.



To provide a direct comparison with similarity-based measures, a representative signal pair was additionally evaluated under aligned and mismatch conditions using identical underlying data. Under correct alignment, correlation and coherence indicate strong similarity together with measurable circR coordination. Under mismatch pairing, correlation and coherence remain comparably elevated, whereas circR collapses sharply, indicating loss of alignment-dependent structure.

This establishes a critical distinction: conventional similarity measures may remain elevated under constraint violation, whereas the proposed observable is accepted only when it systematically degrades under falsification. The framework therefore evaluates not persistence of similarity, but dependence on the defining structural constraint itself.

Across all evaluations:

- the observable remains well-defined and reproducible,
- mismatch pairing acts as a decisive falsifier,
- PASS and FAIL outcomes arise naturally under the fixed contract,
- and adversarial or unrelated pairings are consistently rejected.

Importantly, these results do not imply causality, physical mechanism, or predictive capability. The analysis is restricted to evaluation of alignment-dependent coordination under a frozen falsification protocol.

2.4 n_4 — Closure Structure (Ramsey)

The highest-level evaluation applies the n_4 detector to test for **closure structure** in paired representations derived from real experimental systems. Unlike n_1 , which evaluates directional asymmetry, and n_2 , which evaluates alignment-dependent coordination, the n_4 detector evaluates whether a paired system satisfies a stricter condition of internally consistent structural closure under controlled perturbation.

The primary evaluation uses Ramsey coherence measurements, which provide time-resolved experimental signals across multiple qubit-pair configurations. Each dataset is transformed into paired representations:

- (U_E) (electric-like component),
- (U_M) (magnetic-like component),

forming the closure representation used for n_4 evaluation. The detector operates under a frozen structural fragility contract using the observable:

$$C_4 = \log \left(\frac{U_E + \epsilon}{U_M + \epsilon} \right)$$

with the headline metric defined by the median closure deviation ($|C_4|$). Lower values indicate stronger closure consistency, whereas higher values indicate structural imbalance.

Closure is accepted only when:

- real pairing exhibits low deviation,
- and both mandatory controls produce increased deviation relative to the real pairing.

The mandatory controls are:

- **gauss_zero**: destroys structure through noise perturbation,
- **em_swap**: disrupts the internal pairing structure.

Under the frozen directional rule, a PASS outcome is assigned only when both controls increase closure deviation relative to the real condition. No thresholds, tuning, or optimization are introduced.

A controlled synthetic reference case first confirms detector behavior under ideal conditions. Under correct pairing, the real configuration exhibits minimal closure deviation, whereas both perturbation controls systematically increase deviation, satisfying the directional closure rule exactly. This validates detector correctness under controlled conditions rather than demonstrating real-world closure.

The detector was then applied to Ramsey coherence datasets across multiple qubit-pair configurations. Results exhibit selective PASS and FAIL behavior across pairings:

- internally consistent pairings satisfy the directional degradation rule,
- whereas inconsistent or mismatched pairings fail under the frozen contract.

Representative examples include:

- pair (0+1) → PASS,
- pair (0+3) → FAIL.

Across evaluated configurations:

- total pairs = 8,
- PASS = 6,
- FAIL = 2.

This establishes that closure is not universally present across all pairings, but depends on internally consistent structural organization. PASS cases satisfy degradation under both mandatory controls, whereas FAIL cases are explicitly retained when the directional rule is not satisfied.

Importantly, low closure deviation alone is insufficient for acceptance. Closure fails whenever:

- controls do not increase deviation,
- pairing is internally inconsistent,
- or apparent structure arises from noise, drift, or mismatch.

Adversarial evaluations, including cross-paired Ramsey combinations, unrelated datasets, and oscillatory but mispaired signals, consistently produce FAIL outcomes, confirming that closure cannot be reproduced through superficial balance or generic synchrony alone.

Across all evaluations:

- the detector remains deterministic and reproducible,
- PASS and FAIL outcomes arise naturally under the frozen contract,
- and closure is accepted only when degradation under both controls is preserved.

These results establish closure as a selective and pair-dependent structural property rather than a generic consequence of coordination or oscillatory behavior. Importantly, the analysis does not imply physical closure mechanisms, predictive capability, or quantum interpretation. It is restricted to falsifiable evaluation of pair-dependent structural consistency under a fixed validation protocol.

2.5 Cross-Domain Discriminative Behavior

To evaluate whether the n_2 detector exhibits consistent and selective behavior across heterogeneous systems, the interaction-level observable was applied across industrial, environmental, and grid-scale datasets under a fully frozen evaluation contract. The objective of this analysis is not to establish universality, but to determine whether interaction detection remains structurally grounded and falsifiable across unrelated domains.

All datasets were evaluated using an identical protocol:

- observable: circR,
- mandatory falsifiers: phase randomization, block shuffle, and mismatch pairing,
- deterministic execution conditions,
- and no parameter tuning across datasets.

Under this fixed pipeline, the detector produces distinct outcomes across domains:

- Bearing systems → FAIL,
- Wind systems → PASS_STRONG,
- Grid systems → PASS_STRONG.

Importantly, these outcomes are not determined by observable magnitude alone. Systems exhibiting elevated similarity may still fail, whereas weaker signals may satisfy the falsification criterion when alignment dependence is preserved.

The decisive condition across all domains is the response under mismatch pairing:

Table 1. n_2 interaction outcome under mismatch-controlled falsification.

Condition	Outcome
no mismatch collapse	FAIL
collapse under mismatch	PASS_STRONG

This establishes the central operational principle of the detector: **interaction is defined by alignment dependence rather than by observable magnitude or generic similarity.**

Domain-specific behavior further illustrates this distinction:

- **Bearing systems:** similarity without alignment-dependent interaction → FAIL,
- **Wind systems:** strong alignment-dependent coordination → PASS_STRONG,
- **Grid systems:** weak but consistent coordination with mismatch collapse → PASS_STRONG.

The grid results are particularly important because they demonstrate that interaction can remain structurally valid even at relatively small circR magnitudes, provided that the observable degrades correctly under falsification. Conversely, the bearing results demonstrate that elevated similarity alone is insufficient for structural acceptance when mismatch collapse is absent.

Across all evaluated systems:

- PASS occurs only when structure collapses under mismatch pairing,
- FAIL occurs when apparent structure persists under disruption,
- and intermediate magnitudes do not produce false-positive interaction claims.

Interaction in wind and grid systems satisfies the full n_2 criterion through clear collapse under mismatch pairing, confirming alignment-dependent structure under the frozen detector contract.

These results demonstrate that the detector behaves consistently across unrelated domains without adaptive tuning or domain-specific optimization. The same fixed evaluation logic produces domain-specific outcomes governed by falsification response rather than by raw similarity or coordination strength alone.

2.6 Cross-Level Interpretation

The combined results across n_1 , n_2 , and n_4 establish a hierarchical progression of increasingly strict structural constraints:

Table 2. Structural meaning of detector levels.

Detector Level	Structural Meaning
n_1	ordering sensitivity
n_2	alignment-dependent interaction
n_4	pair-dependent closure

This progression defines the operational hierarchy:

time-order → interaction → closure

with each detector level enforcing a stronger falsification requirement than the preceding level.

For n_1 , directional asymmetry is observable under real conditions, but rarity under all required nulls is not satisfied, resulting in conservative NO_CLEAR classifications. This demonstrates that ordering sensitivity alone is insufficient for validated time-order structure.

For n_2 , measurable coordination is accepted only when the observable collapses under mismatch pairing. Wind and grid systems satisfy this criterion, whereas bearing systems fail despite measurable similarity, confirming that interaction requires alignment dependence rather than persistence of coordination alone.

For n_4 , closure is accepted only when internally consistent pairing remains stable under real conditions and degrades under both mandatory perturbation controls. Mixed PASS / FAIL outcomes across Ramsey pairings demonstrate that closure represents a stricter and more selective structural condition than interaction itself.

The integrated cross-domain behavior is summarized below:

Table 3. Cross-domain validation summary across detector levels.

Domain	Detector	Behavior	Falsifier Response	Verdict	Meaning
GRB	n_1	directional asymmetry observed	reproduced under block-shuffle null	NO_CLEAR	not uniquely time-order dependent
GOES	n_2	alignment-dependent coordination	collapses under mismatch pairing	PASS_STRONG	interaction depends on preserved alignment
Ramsey	n_4	closure consistency under pairing	degraded under independent controls	Mixed (PASS / FAIL)	selective closure consistency

Each detector level therefore introduces a stronger falsification requirement:

Table 4. Progressive falsification requirements across detector levels.

Level	Defining Requirement
n_1	fails under ordering destruction
n_2	collapses under mismatch pairing
n_4	degrades under multiple independent controls

The resulting principle is consistent across all evaluations:
structure is accepted only when it fails correctly under its defining falsifier.

Importantly, the hierarchy does not imply replacement between detector levels. Each detector evaluates a distinct structural property:

- $n_1 \rightarrow$ directional sensitivity,
- $n_2 \rightarrow$ relational interaction,
- $n_4 \rightarrow$ internally consistent closure.

Together, the detectors separate different forms of structure while preventing reduction of structural interpretation to a single similarity metric. In particular, the results demonstrate that closure is stricter than interaction: n_2 detects alignment-dependent coordination, whereas n_4 additionally requires internally consistent pairing that degrades correctly under independent perturbation controls.

2.7 Baseline Comparison and Controlled Illustration

To contextualize the behavior of the proposed observables under controlled conditions, a comparative evaluation was performed using the bearing dataset under identical preprocessing, windowing, and evaluation conditions for all methods. The purpose of this comparison is not to establish superiority, but to clarify differences in structural sensitivity between conventional similarity-based measures and the falsification-based framework.

The following measures were evaluated:

- **correlation** → amplitude similarity,
- **coherence** → spectral similarity,
- **circR** → phase-aligned coordination.

All methods exhibit strong separation between evaluated conditions with large effect sizes and statistically significant results. However, the direction of separation differs systematically across observables:

- correlation and coherence increase under fault conditions,
- whereas circR is elevated in the structurally aligned normal regime.

This distinction reflects differences in the structural properties captured by each method rather than instability or noise sensitivity. Conventional measures primarily capture amplitude or spectral similarity, whereas circR evaluates alignment-dependent phase coordination.

Importantly, this comparison demonstrates complementarity rather than competition between methods. Conventional similarity metrics remain valid within their intended domains, while the proposed framework evaluates a different criterion: whether detected structure depends on a defining constraint and degrades correctly when that constraint is violated.

This distinction becomes stricter at the n_4 level, where closure is accepted only when internally consistent pairing degrades under mandatory structural falsifiers rather than merely exhibiting coherence or synchrony.

To further illustrate the distinction between linear similarity and phase-dependent structure, a controlled synthetic example was constructed in which phase alignment is preserved while amplitude correlation is minimized. The signals were designed to:

- preserve phase structure,
- minimize linear correlation,
- and avoid external coupling.

Under these conditions:

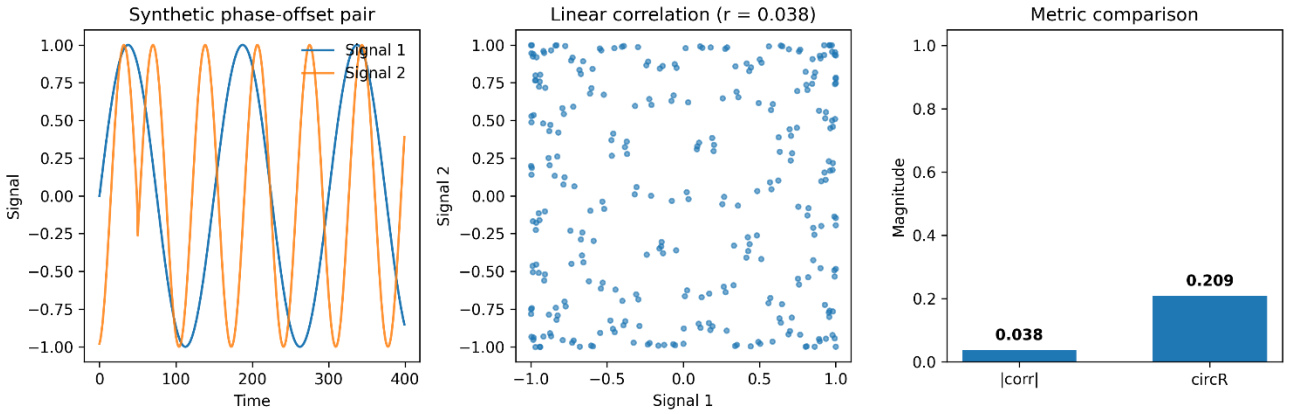
- correlation remains near zero,

- whereas circR remains stable and non-zero.

As shown in **Figure 4**, phase-aligned structure can therefore remain detectable even when conventional linear correlation is weak. The example is included strictly as a controlled behavioral illustration and not as evidence of real-world interaction.

Figure 4. Synthetic illustration of phase-aligned structure without strong linear correlation.

A controlled signal pair exhibits phase-aligned coordination while maintaining weak linear dependence. Linear correlation remains near zero, whereas the circR observable remains non-zero, demonstrating sensitivity to phase-dependent coordination not captured by correlation alone. This example is included strictly as a controlled illustration of detector behavior rather than as evidence of real-world structure.



Across these comparisons, a consistent pattern emerges:

- conventional similarity measures may remain elevated under constraint disruption,
- whereas the proposed framework accepts structure only when it degrades under its defining falsifier.

Accordingly, the framework does not infer structure from persistence of similarity alone, but from observable dependence on the preserved structural constraint itself.

Across all real-data evaluations, the framework consistently preserves fail-closed behavior:

- directional patterns that are not statistically rare remain NO_CLEAR,
- weak or null-dominated systems do not produce spurious PASS outcomes,
- and closure is accepted only when internally consistent pairing degrades correctly under mandatory controls.

These results establish falsification-driven degradation, rather than observable magnitude alone, as the operational criterion for structural validity across domains.

3. Discussion

3.1 Core Interpretation

The results across all detector levels form a consistent falsification-based hierarchy rather than a collection of independent observations. Each detector evaluates a distinct structural property under a specific falsification rule, producing the operational progression:

time-order → interaction → closure

Within this hierarchy:

- **n₁** evaluates directional time-order sensitivity,
- **n₂** evaluates alignment-dependent coordination,
- **n₄** evaluates pair-dependent closure consistency under perturbation.

The progression reflects increasing structural constraint and increasing selectivity. At each level, observed behavior is accepted only when it degrades under the falsifier corresponding to its defining structural condition. Structural interpretation is therefore determined not by observable magnitude alone, but by reproducible behavior under controlled disruption.

A central property of the framework is its strict selectivity. Across multiple datasets, many evaluated signals produce **NO_CLEAR** or **FAIL** outcomes. These results are not treated as weaknesses of the framework, but as meaningful outcomes indicating insufficient structural support under falsification. This behavior is observed consistently across detector levels:

- **n₁** detects directional sensitivity but frequently remains **NO_CLEAR** when rarity under all required nulls is not satisfied,
- **n₂** accepts interaction only when coordination collapses under mismatch pairing,
- **n₄** imposes the strictest requirement, accepting closure only when internally consistent pairing degrades under multiple independent controls.

As a result, higher detector levels impose progressively stronger constraints, and positive outcomes represent increasingly selective forms of structure.

A unifying principle emerges across all evaluations:

structure is accepted only when it degrades under its defining falsifier.

At the n_1 level, surrogate nulls and reversal modify directional statistics, but classification depends on rarity under the required null families. At the n_2 level, interaction must collapse under mismatch pairing. At the n_4 level, closure must degrade under both mandatory perturbation controls. This consistent falsification-driven behavior ensures that structural acceptance depends on reproducible constraint sensitivity rather than persistence of similarity, smoothness, or synchrony.

The framework additionally enforces a strict separation between detection and interpretation. Each detector makes only a limited operational claim:

- n_1 indicates time-order contribution only when supported under surrogate rarity,
- n_2 identifies interaction only when alignment dependence is preserved,
- n_4 identifies closure only when internally consistent pairing degrades correctly under perturbation.

None of these outcomes imply causality, predictive capability, or physical mechanism. Conclusions remain explicitly bounded by observable behavior under falsification, preventing escalation of statistical behavior into unsupported mechanistic interpretation.

Cross-domain application further demonstrates consistent discriminative behavior under a unified protocol. The same frozen evaluation rules produce distinct outcomes across unrelated systems:

- GRB signals exhibit directional asymmetry but remain NO_CLEAR,
- GOES signals exhibit alignment-dependent interaction and produce PASS_STRONG outcomes,
- Ramsey coherence datasets exhibit selective closure with mixed PASS / FAIL behavior,
- bearing systems fail despite measurable similarity,
- wind and grid systems satisfy the interaction criterion through mismatch collapse.

These outcomes are governed by falsification response rather than observable magnitude alone, demonstrating that the framework acts as a selective discriminator of structurally supported behavior rather than as a generic pattern detector.

The integrated behavior across detector levels establishes a progressively constrained interpretation of structure:

- n_1 identifies directional sensitivity without promoting it to irreversibility,
- n_2 identifies interaction while rejecting similarity-based artifacts,
- n_4 identifies closure selectively under internally consistent pairing constraints.

Importantly, both positive and negative outcomes are explicitly preserved throughout the framework.

The resulting operational insight is therefore:

meaningful structure is defined by reproducible degradation under controlled falsification rather than by apparent persistence or signal magnitude alone.

This perspective suggests a structured workflow for time-series evaluation:

1. evaluate directional ordering sensitivity (n_1),
2. evaluate alignment-dependent interaction (n_2),
3. evaluate closure under perturbation (n_4).

At each stage, conclusions are accepted only when supported by the corresponding falsification criteria, while weak or ambiguous patterns remain rejected or classified conservatively as NO_CLEAR.

The framework is therefore best understood as a conservative, falsification-based validation system for structured dependencies in time-series data. It is not intended as a universal detector, predictive model, or replacement for existing analytical methods. Its primary contribution is the reproducible and selective validation of structure under explicit falsification constraints.

In particular, the contribution of n_4 lies primarily in controlled rejection. Generic oscillation, synchrony, or smooth decay is insufficient for closure. A positive n_4 outcome requires that real pairing remain more structurally closed than both mandatory perturbation controls, while FAIL outcomes are explicitly preserved whenever this condition is not satisfied. The n_4 detector therefore functions as a pair-dependent structural audit layer rather than as a generic synchrony or coherence measure.

3.2 Relationship to Conventional Measures

Comparison with conventional methods highlights a complementary rather than competitive relationship between the proposed framework and existing signal-analysis approaches. Conventional measures and the proposed observables evaluate different structural properties:

- **correlation** primarily captures amplitude similarity,
- **coherence** captures spectral similarity,
- **circR** evaluates phase-aligned coordination under alignment dependence.

Across evaluated datasets, all methods may exhibit statistically significant separation while differing systematically in the direction and interpretation of that separation. This demonstrates that phase-sensitive interaction is not equivalent to amplitude-based or spectral similarity.

A particularly important distinction is that conventional similarity measures may remain elevated even when structural constraints are disrupted. Under mismatch pairing, correlation

and coherence can preserve substantial similarity despite destruction of the defining alignment relationship. In contrast, the proposed framework accepts interaction only when the observable degrades under its defining falsifier. Structural validity therefore depends on falsification-driven collapse rather than persistence of similarity alone.

This distinction becomes increasingly strict at higher detector levels. While n_2 evaluates alignment-dependent coordination, n_4 additionally requires internally consistent pairing together with degradation under multiple independent perturbation controls. Closure is therefore not inferred from coherence, smooth decay, or oscillatory behavior alone.

The controlled synthetic illustration further clarifies this distinction. In a constructed example with minimal linear correlation but preserved phase alignment, circR remains stable and non-zero while correlation remains near zero. This demonstrates that phase-dependent structure can exist independently of amplitude similarity and that the proposed observable captures a distinct structural perspective rather than a reformulation of conventional correlation metrics.

Accordingly, the framework should not be interpreted as a replacement for established statistical or signal-processing methods. Instead, it provides a complementary validation layer that evaluates whether observed structure depends on a defining constraint and degrades appropriately when that constraint is violated. This allows the framework to reduce false-positive structural interpretation while remaining compatible with broader analytical approaches.

3.3 Limitations

The framework is intentionally conservative and operates as an operational validation system rather than a predictive or mechanistic model. Its purpose is to evaluate whether observed structure depends on explicitly defined constraints and degrades correctly under falsification, not to infer physical laws, causal mechanisms, or future behavior.

Accordingly, the framework does not provide:

- prediction or forecasting capability,
- causal inference,
- trajectory extrapolation,
- or mechanistic explanation of observed structure.

The detectors evaluate only observable behavior under fixed falsification conditions. Any apparent stability or sensitivity reflects properties of the observable under the evaluated data and does not imply predictive power or directional influence between signals.

The framework is additionally representation-dependent. Structural outcomes depend on the selected signal representation and pairing construction:

- n_2 evaluates interaction under a specific alignment model,

- n_4 evaluates closure using a fixed $((U_E, U_M))$ pairing representation.

Alternative representations, feature constructions, or pairing schemes may produce different outcomes. The framework therefore does not claim invariance under arbitrary transformations or guarantee detection of all forms of structure present within a signal. Similarly, the evaluated datasets represent limited domains and should not be interpreted as exhaustive demonstrations of all possible structural regimes.

Importantly, detection of structure does not imply underlying mechanism. The framework identifies:

- directional temporal organization (n_1),
- alignment-dependent coordination (n_2),
- and pair-dependent closure consistency under perturbation (n_4),

but does not determine how such behavior is generated or whether the observed structure reflects fundamental or emergent processes. Multiple distinct mechanisms may produce similar observable behavior under the evaluation rules.

All outcomes are interpreted under a strict fail-closed principle:

- PASS requires satisfaction of all validation conditions,
- FAIL is assigned when falsification conditions are violated,
- NO_CLEAR is retained when evidence is insufficient.

A substantial proportion of evaluated cases produce NO_CLEAR or FAIL outcomes. This reflects deliberate selectivity rather than lack of sensitivity. The framework is intentionally designed to minimize false-positive structural interpretation and does not guarantee positive detection.

The framework should therefore be understood as a complementary validation layer for structure assessment under falsification rather than as a replacement for statistical modeling, machine learning, or domain-specific analytical methods. It prioritizes conservative and reproducible validation over unconstrained applicability or maximal detection sensitivity.

3.4 Reproducibility and Determinism

All evaluations are conducted under frozen detector contracts in which observables, falsifiers, null families, and decision rules are fixed prior to analysis and are not modified post hoc. Execution is deterministic under fixed inputs, seeds, and parameters:

- identical inputs produce identical outputs,
- logic fingerprints remain constant,
- and classifications are reproducible without variation.

Independent audit tools reproduce detector outcomes without altering decision logic and preserve both positive and negative results. Complete evidence packs, including observable summaries, falsification outputs, audit records, and logic fingerprints, are generated within the evaluation pipeline and can be reproduced under identical configurations.

3.5 Final Conclusion

This work presents a falsification-first, multi-level framework for validating structure in time-series data through three progressively constrained detector levels:

- n_1 — time-order structure,
- n_2 — interaction structure,
- n_4 — closure structure.

Each detector operates under a fixed evaluation contract consisting of predefined observables, mandatory falsifiers, and deterministic decision logic. Across all levels, structure is interpreted not by observable magnitude alone, but by reproducible degradation under controlled constraint violation.

Application across multiple domains demonstrates consistent and interpretable behavior under the frozen framework:

- n_1 detects directional sensitivity while conservatively rejecting unsupported irreversibility claims,
- n_2 identifies alignment-dependent coordination through mismatch collapse,
- n_4 identifies selective closure only when internally consistent pairing degrades under mandatory perturbation controls.

Across all evaluations, both positive and negative outcomes are explicitly preserved.

The primary contribution of the framework is not increased detection sensitivity, but a reproducible mechanism for rejecting unsupported structure. The results establish a consistent operational principle:

meaningful structure is accepted only when it degrades under its defining falsifier.

Within this interpretation, the hierarchy:

time-order → interaction → closure

should be understood as a system of progressively stricter validation filters rather than a hierarchy of structural strength. Each level imposes additional falsification constraints, ensuring that increasingly selective forms of structure are retained only when supported under controlled perturbation.

The framework is therefore best understood as a conservative and falsification-based validation system for structured dependencies in time-series data rather than as a predictive, generative, or universal detection model. In particular, the contribution of n_4 lies in controlled structural rejection: generic synchrony, oscillation, or smooth decay is insufficient for closure, and closure is accepted only when pair-dependent structure degrades correctly under both mandatory perturbation controls while FAIL outcomes are explicitly preserved.

Taken together, the results demonstrate that structure validation can be made falsifiable, reproducible, and multi-level across heterogeneous systems while preserving strict operational boundaries and conservative interpretation.

4. Methods

4.1 Framework Overview

The proposed framework consists of a set of frozen detectors designed to evaluate distinct forms of structure in time-series data under explicit falsification constraints. The framework is organized into three detector levels:

- **n_1 (time-order):** evaluates directional time-order asymmetry in a single signal under surrogate-based falsification;
- **n_2 (interaction):** evaluates alignment-dependent coordination between paired signals;
- **n_4 (closure):** evaluates pair-dependent structural validity under controlled perturbation.

Each detector operates under a fixed evaluation contract consisting of:

- a predefined observable,
- mandatory falsifiers or controls,
- and a deterministic decision rule.

Structure is therefore interpreted not by observable magnitude alone, but by its behavior under controlled disruption. A signal is accepted as structurally meaningful only when the defining observable weakens or degrades under violation of its defining constraint.

The detectors are intentionally conservative, selective, and domain-agnostic. No parameter retuning is performed across datasets, and detector logic remains fixed throughout all evaluations. As a result, the framework does not guarantee positive detection and instead preserves FAIL and NO_CLEAR outcomes whenever falsification support is incomplete.

The detector hierarchy introduces progressively stricter structural constraints:

- n_1 requires directional sensitivity under ordering destruction,
- n_2 requires collapse under mismatch pairing,
- n_4 requires internally consistent pairing together with directional degradation under multiple independent controls.

This progression is summarized operationally as:

time-order → interaction → closure

Higher detector levels therefore represent increasingly constrained validation filters rather than increasing signal magnitude or importance. In particular, n_4 represents the strictest level, rejecting not only weak patterns but also generic synchrony or coordination that does not satisfy internal structural consistency.

Supporting audit tools operate independently of the detectors and provide secondary verification without altering observables, falsifiers, or decision logic. These audit layers enforce reproducibility, consistency, and evidence integrity across repeated execution conditions.

The framework should therefore be understood as a falsification-first validation system rather than as a predictive model or universal structure detector. Its central operational principle is that meaningful structure must be defined through reproducible degradation under controlled destruction rather than inferred from descriptive statistics alone.

4.2 Detector Definitions and Decision Rules

n_1 — Time-Order Structure

The n_1 detector evaluates whether a single time series exhibits directional time-order asymmetry under surrogate-based falsification. The detector employs a fixed directional statistic summarized under a frozen null-audit protocol. Directional behavior is evaluated relative to mandatory ordering-destroying transformations including:

- phase-randomized surrogates,
- block-shuffled surrogates,
- and time reversal (diagnostic only).

Final classification depends on rarity relative to the surrogate null distributions rather than reversal behavior alone. The detector produces two outcomes:

- **PASS:** directional asymmetry is statistically rare under the required null families;
- **NO_CLEAR:** directional behavior is observed but does not satisfy the required rarity criteria.

The detector does not establish causality, prediction, or irreversibility. It evaluates only whether directional asymmetry depends on preserved temporal ordering under controlled falsification.

n₂ — Interaction Structure

The n₂ detector evaluates alignment-dependent coordination between paired signals using the fixed observable circR, a bounded phase-based coordination statistic. Interaction is accepted only when observed coordination depends on correct temporal alignment and collapses under mismatch pairing.

Mandatory falsifiers include:

- mismatch pairing (decisive),
- phase randomization,
- block shuffle.

The detector produces two outcomes:

- **PASS_STRONG:** coordination is present under real pairing, collapses under mismatch, and is not preserved under surrogate nulls;
- **FAIL:** interaction persists under mismatch or cannot be distinguished from null behavior.

The mismatch falsifier is decisive: any interaction surviving mismatch is rejected. The detector therefore evaluates alignment-dependent coordination rather than generic similarity, coherence, or shared trends.

In this work, the term *coordination* is used operationally to denote interaction that collapses under mismatch pairing without implying causality or physical coupling.

n₄ — Closure Structure

The n₄ detector evaluates pair-dependent structural validity using paired representations ((U_E, U_M)) derived from the original data through a fixed preprocessing pipeline. Closure is defined operationally as internally consistent pairing that remains stable under real conditions and degrades under mandatory perturbation controls.

The closure observable is:

$$C_4 = \log \left(\frac{U_E + \epsilon}{U_M + \epsilon} \right)$$

summarized using the median absolute closure deviation:

$$\text{abs_}C_{4,\text{median}} = \text{median}_w |C_4|$$

Lower values indicate stronger closure consistency, whereas higher values indicate structural imbalance.

Mandatory perturbation controls include:

- gauss_zero,
- em_swap.

Closure is accepted only when real pairing exhibits lower closure deviation than all mandatory controls. The detector produces two outcomes:

- **PASS_directional_controls:** real pairing shows consistently lower deviation than controls;
- **FAIL:** degradation under controls is not preserved.

No tuned thresholds are used; acceptance depends only on directional comparison under the frozen contract.

The detector does not establish physical laws, conservation principles, or causal mechanisms. It evaluates only whether structure depends on internally consistent pairing and degrades correctly under controlled perturbation.

4.3 Ladder Interpretation and Validation Logic

The three detectors defined in this framework form a hierarchical ladder of progressively stricter structural constraints. Each detector evaluates a distinct form of structure under its own falsification rule, producing the operational progression:

time-order → interaction → closure

The detectors are evaluated independently and do not require sequential passing across levels. Instead, the hierarchy should be interpreted as a system of increasingly restrictive validation filters rather than as a ranking of structural strength.

At the lowest level, n_1 evaluates whether a signal depends on temporal direction. The defining requirement is directional asymmetry that degrades under ordering-destroying null transformations. This level evaluates ordering sensitivity without imposing relational or pairing constraints.

The n_2 detector introduces a stronger relational condition. Structure must now depend on correct alignment between paired signals and collapse under mismatch pairing. This transition represents a shift from ordering sensitivity to alignment-dependent coordination.

The n_4 detector imposes the strictest requirement. Structure must depend on internally consistent pairing and degrade correctly under multiple independent perturbation controls. Closure is therefore not inferred from generic coordination or synchrony alone, but only from pairing-dependent structural consistency under falsification.

Each detector level introduces a stronger falsification condition:

Table 5. Progressive falsification requirements across detector levels.

Detector Level	Defining Structural Requirement
n_1	collapse under ordering destruction
n_2	collapse under mismatch pairing
n_4	degradation under multiple perturbation controls

As a result, higher detector levels are more selective and more constrained. A positive n_4 outcome therefore represents stronger falsification support than a positive n_2 or n_1 outcome, not simply a larger observable magnitude.

The ladder structure additionally separates different forms of structure that are often conflated under generic similarity measures:

- n_1 isolates directional ordering sensitivity,
- n_2 isolates alignment-dependent interaction,
- n_4 isolates internally consistent closure structure.

This separation reduces interpretive ambiguity and prevents structurally distinct behaviors from collapsing into a single similarity metric.

The resulting operational principle is consistent across all detector levels:

structure is accepted only when it degrades under its defining falsifier.

This falsification-driven interpretation forms the conceptual backbone of the framework and ensures that structural validation remains reproducible, selective, and resistant to overinterpretation.

4.4 Supporting Audit Layers and Reproducibility

Each detector is paired with supporting audit tools that provide independent verification without altering detector logic, observables, falsifiers, or decision rules. These audit layers operate strictly as secondary validation systems and are included to enforce reproducibility, evidence integrity, and implementation consistency across runs.

The audit systems include:

- **n₁**: falsification ledgers that preserve complete surrogate evaluation records and retain both PASS and NO_CLEAR outcomes;
- **n₂**: FAR-matched evaluators that verify PASS_STRONG and FAIL behavior under identical evaluation conditions;
- **n₄**: closure audit tools that independently recompute closure behavior under the frozen contract while preserving both PASS and FAIL outcomes.

These tools do not modify detector behavior or reinterpret outcomes. Their role is strictly to confirm that results remain stable, reproducible, and traceable under deterministic execution conditions.

All evaluations are conducted under frozen contracts:

- observables are fixed prior to analysis,
- falsifiers and null families are predefined,
- decision rules are immutable,
- and no post hoc retuning is permitted.

Execution remains deterministic under fixed inputs, seeds, and parameters:

- identical inputs produce identical outputs,
- logic fingerprints remain stable,
- and complete evidence packs can be reproduced exactly under identical configurations.

This separation between detector logic and audit verification is essential to the framework design. It ensures that outcomes are both operationally defined through fixed contracts and independently verifiable through reproducible audit layers.

4.5 Positioning and Claim Boundaries

The framework introduced in this work is a falsification-based validation system for evaluating structure under explicit operational constraints. Each detector addresses a narrowly defined structural question:

- **n₁** evaluates directional time-order asymmetry under surrogate testing;
- **n₂** evaluates whether interaction depends on correct temporal alignment;
- **n₄** evaluates whether closure depends on internally consistent pairing and controlled perturbation response.

These evaluations establish the presence or absence of structure only within the scope of the frozen detector contracts.

The framework does not:

- provide forecasting or predictive capability,
- establish causal relationships,
- infer governing physical laws,
- or claim universal detection of all structure types.

All conclusions are intentionally restricted to observable behavior under controlled falsification.

The framework is additionally not intended to replace conventional statistical analysis, machine learning systems, or domain-specific analytical methods. Instead, it functions as a complementary validation layer for determining whether observed structure remains meaningful under controlled disruption.

This restriction is deliberate. By limiting claims to falsifiable operational behavior, the framework prioritizes:

- reproducibility,
- interpretive clarity,
- resistance to hidden tuning,
- and protection against overinterpretation.

Within this design, FAIL and NO_CLEAR outcomes are treated as meaningful and informative results rather than as detector deficiencies.

Data Availability

The datasets analyzed in this study, together with derived evaluation outputs and audit artifacts, are available from the corresponding author upon reasonable request. Publicly accessible datasets used in this work are referenced within the Supplementary Information. All evaluations were performed under frozen detector contracts without dataset-specific retuning.

Code Availability

The analysis code implementing the n_1 , n_2 , and n_4 detectors, together with the evaluation pipeline, falsification controls, and audit procedures, is available from the corresponding author and will be made publicly accessible upon publication. All detector contracts, observables, falsifiers, and decision rules are fixed prior to evaluation and remain unchanged across datasets and runs.

Author Contributions

The author conceived the framework, designed the validation methodology, implemented the evaluation and audit pipeline, performed the computational experiments, analyzed the results, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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AI Usage Declaration

Artificial intelligence tools were used only for language refinement, formatting assistance, and organizational support. All scientific concepts, methodological structures, analyses, and conclusions were developed and verified by the author.

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